Lesson 14 Teacher's Script

Raising the Stakes

VIDEO GAME CREATION

[0:00-0:05] Welcome Back

"Welcome back! Your game moves, shoots, sounds good, and reacts to hits. Today, it starts to feel like a REAL arcade game - one that gets harder the longer you survive."

"Some of you may remember a hint from Lesson 11's Teacher's Key, or a comment in the TIC-80 or Scratch materials, that hard-coding 5 enemies per wave was 'a deliberate simplification.' Today's the day that pays off."

[0:05-0:15] The Foreshadowed Feature: One Formula, Two Places

"Here's the whole idea in one sentence: instead of always spawning 5 enemies, we spawn (4 + wave) enemies. Wave 1: still 5. Wave 2: 6. Wave 3: 7. One small formula, and your game already escalates."

"The catch: that formula has to appear in TWO places that already exist in your code - wherever you spawn enemies, AND wherever enemies_remaining gets reset for a new wave. If those two disagree, your win-check breaks. This is exactly the kind of 'one source of truth' thinking we've been building since Lesson 10's reusable reset function."

[Timing checkpoint: wrap by 0:15]

[0:15-0:35] Adding Speed to the Mix

[Demo cue: switch to GameMaker, open a project with Lesson 13's flash and explosion working]

"Count alone is a good start, but speed makes it feel dangerous. Watch: in spawn_enemies(), right after instance_create_layer() makes each enemy, I set inst.enemy_speed = 1 + (wave - 1) * 0.5. Wave 1 stays exactly the same. Wave 3 is noticeably faster." [Build it live, demonstrate wave 1 versus wave 3 side by side if time allows]

"That new enemy_speed variable has to actually get used, so flip over to obj_enemy's Step event: it currently just says y += 1. Change that one line to y += enemy_speed, and suddenly the number we set at spawn time is doing real work every frame. Notice what did NOT change: the rest of the Step event, from Lesson 4. It already ran every frame - today we just made one of its numbers depend on the wave."

"TIC-80 folks: your engine doesn't have a per-instance speed variable yet, since Lesson 7 only offered that as an OPTIONAL idea. Today it becomes required - you're adding it for real, if you haven't already."

[Timing checkpoint: wrap by 0:35]

[0:35-0:45] This Week's Twist: Student's Choice

"Level 3 works differently this week. Instead of 2-3 small stretch ideas, you're choosing ONE feature from a list of five real capstone-level ideas: destructible shields, a power-up, a bonus enemy, formation movement, or a title/game-over screen."

"These are open-ended on purpose. A good-faith attempt at the general idea - even if it's rough - is a complete Level 3 finish today. Don't feel like you need to polish it. Pick whichever one sounds the most fun to YOU."

[Timing checkpoint: wrap by 0:45]

[0:45-0:55] Today's Exercise & Handouts

"For lab today, open your own project from last week - not a new one. No new assets needed, except 001 Game Creator folks, who are adding two more pre-placed enemy copies in the Map editor first."

[Hand out engine-specific handouts]

"Level 1 and 2 today, then pick your Level 3 feature. Test a full 3-wave round when you're done, and confirm restarting brings everything back to wave 1's original feel."

"Questions before we start lab? ... Let's raise the stakes."

Lab Time Notes

- The most common mistake this week: updating the spawn-count formula in one place but not the enemies_remaining reset in the other - watch for this specifically, it's an easy one to half-do.
- 001 Game Creator students need a few extra minutes up front to correctly add and name enemy_6/enemy_7 in the Map editor before any scripting begins.
- Difficulty scaling can tip into feeling unfair if a student's increments are too aggressive - if wave 3 feels impossible rather than exciting, help them dial back the numbers rather than pushing through.
- Level 3 this week is genuinely open-ended - circulate and help students scope their chosen feature down to something achievable in the remaining lab time, rather than letting anyone get stuck chasing a fully-polished version.
- At 1:50, do a quick share-out - ask a few students to demo their wave 3 and their chosen Level 3 feature, whatever state it's in.